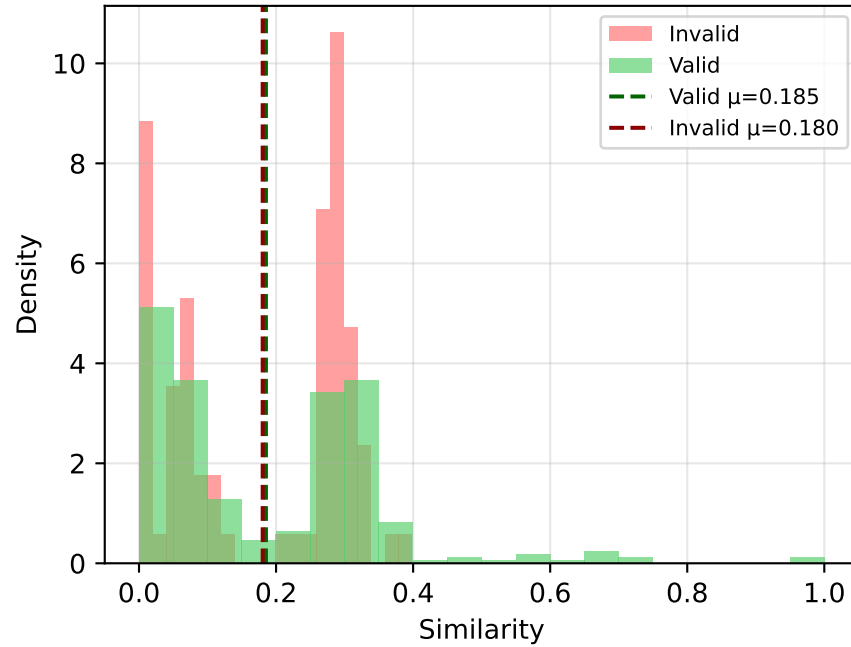
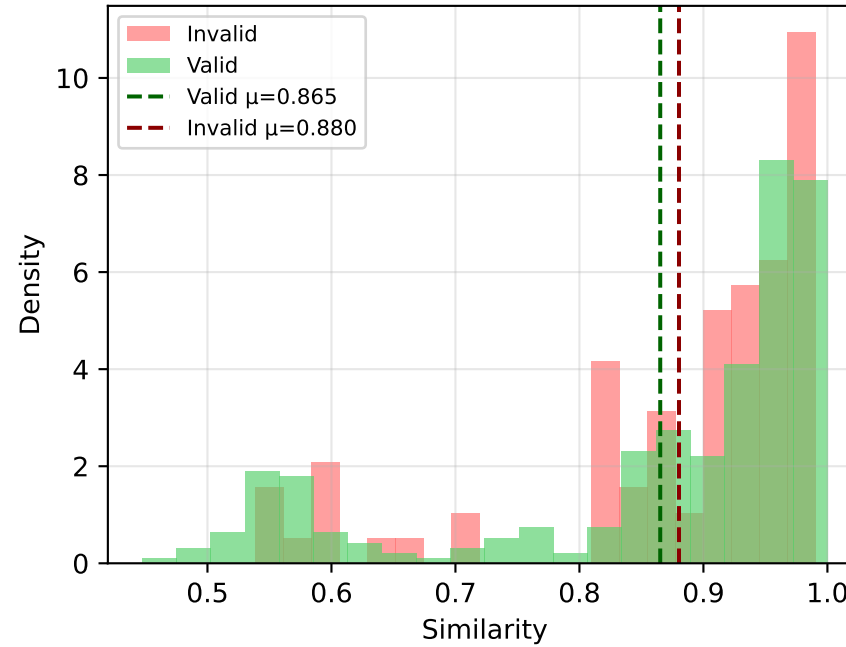


Ablation Study: Similarity Alone Is Insufficient for Relationship Validation

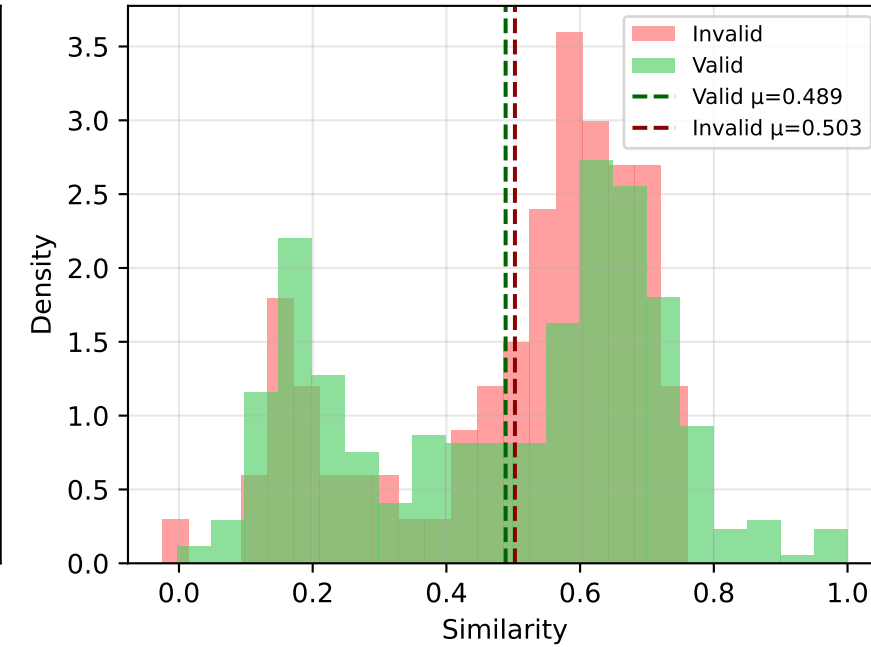
TF-IDF cosine



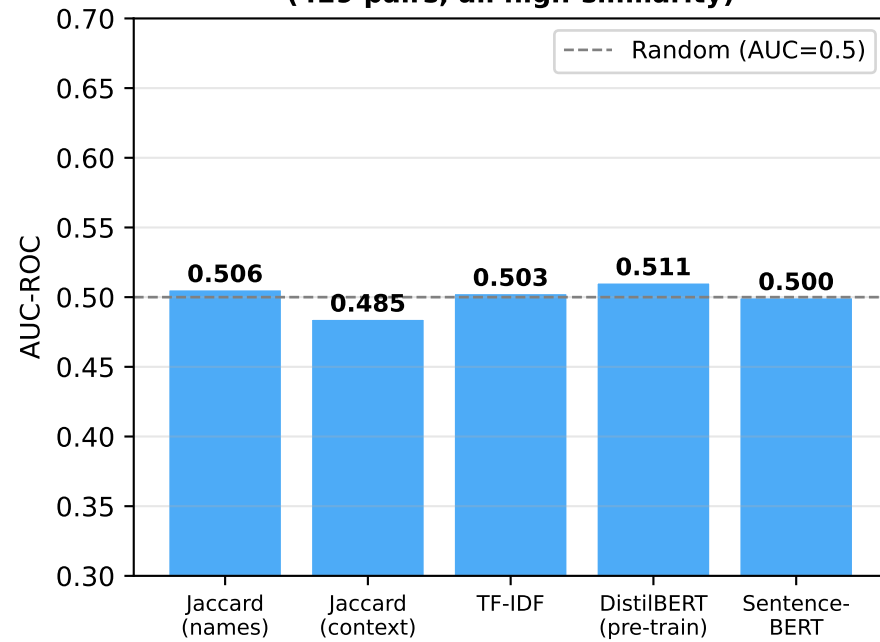
DistilBERT (pre-trained)



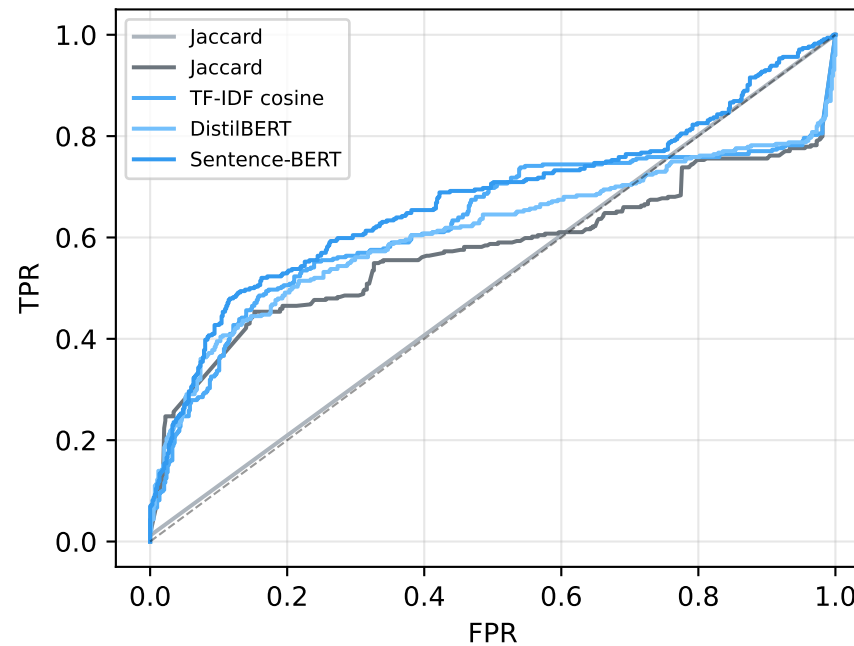
Sentence-BERT (MiniLM)



AUC on Pre-filtered Pairs (429 pairs, all high-similarity)



ROC (full eval set)



Key finding:
On pre-filtered pairs
(all high similarity),
no similarity method
achieves AUC > 0.52.

The LLM semantic
reasoning is what
provides 80.2%
precision.